Dr. Li Hsu

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Research Interests

Decoding the biological roles of the diversity and structural variation of glycans via incorporating chemoenzymatic synthesis and studying molecular recognition and molecular dynamics.

Professional Experience

Guest Scientist

Freie Universität Berlin (FU Berlin) | Berlin, Germany | *15.06.2026 – Present*

- Formulating structural frameworks for biomimetic glycopeptide conjugate research.

01/2024 – 06/2025 | Postdoctoral Research Fellow

Postdoctoral Research Fellow

Genomics Research Center, Academia Sinica | Taipei, Taiwan | *01/01/2024 – 30/06/2025*

Advisor: Prof. Dr. Chi-Huey Wong

- Synthesized complex glycans to systematically interrogate carbohydrate-protein interactions within the Siglec-10/CD24 immune checkpoint.
- Evaluated therapeutic pathways optimizing antibody-dependent cellular phagocytosis (ADCP) mechanisms.
- Developed site-specific single-position functionalization methodologies (fluorination and methylation) to modulate glycan conformational stability and receptor binding kinetics.

Postdoctoral Researcher

Department of Biomedical Science and Engineering, National Central University | Taiwan | *01/05/2022 – 30/04/2023*

Advisor: Prof. Chen-Han Huang

- Designed and engineered targeted functional coatings to enhance biocompatibility profiles for clinical-grade medicinal apparatus (stents and contact lens).
- Synthesized poly(sulfobetaine methacrylate) (PSBMA)-chitosan polymers grafted onto titanium substrates to investigate intra- and intermolecular hydroxyl interactions governing surface bio-fouling.
- Refined zwitterionic grafting process on silicon materials.
- Mentored 2 Master of Science and 5 undergraduate researchers in core synthesis protocols, analytical criteria, and laboratory safety operations.

Senior R&D Engineer

Taiwan User-Friendly Technology Ltd. | Taiwan | *01/06/2021 – 30/04/2022*

- Designed and compiled an ionic liquids and silicon magnetic nanomaterials library for ultra sensitive real-time food allergen sensor production.
- Established and validated kinetic performance models for oxidoreductases (HRP/GOx) electro-biochemical assays.
- Assisted product mechanical designing.

Education

Ph.D. in Chemistry

Department of Chemistry, National Taiwan University | Taipei, Taiwan | *01/09/2012 – 31/01/2021*

Advisor: Prof. Dr. Chi-Huey Wong

- Developed chemo-enzymatic methodologies for systematical chemical glycobiology studies.
- Mapped isomeric processing preferences and of glycosyltransferases.
- Deciphered avian H5N2 hemagglutinin underlying recognition preferences via glycan array.
- Provided calibration standards for MASS glycan sequencing.
- Assisted engineering novel vaccine candidates targeting HIV by supplying glycans for structural fingerprints identification.

B.S. in Chemistry

National Chiao-Tung University | Taiwan | *01/09/2006 – 30/06/2010*

Peer Reviewed Publications

- **Pawar, S.**, **Hsu, L.**, Reddy, T. N., Ravinder, M., Ren, C. T., Lin, Y. W., Cheng, Y. Y., Lin, T. W., Hsu, T. L., Wang, S. K., Wong, C.H., Wu, C. Y. Synthesis of Asymmetric N-Glycans as Common Core Substrates for Structural Diversification through Selective Enzymatic Glycosylation. *ACS Chem. Biol.* **2020**, 15(9), 2382–2394.
  Impact Summary: Established a platform, allowing us to systematically observe whether glycosyltransferases exhibit preferences across glycans' isomeric arms, giving a rare opportunity to differentiate disease from healthy from the side of the glycan production, while providing a library uncovering H5N2 hemagglutinin's branch-biased, context-specific recognition patterns
- Shivatare, S. S., Chang, S. H., Tsai, T. I., Ren, C. T., Chuang, H. Y., **Hsu, L.**, Lin, C. W., Li, S. T., Wu, C. Y., Wong, C. H. Efficient Convergent Synthesis of Bi-, Tri-, and Tetraantennary Complex Type N-Glycans and Their HIV-1 Antigenicity. *J. Am. Chem. Soc.* **2013**, 135(41), 15382–15391.
  Impact Summary: Developed a refined chemoenzymatic approach to construct an extensive complex glycan library. Profiled broadly neutralizing antibodies PG9 and PG16 via microarray assays, identifying a structurally simplified, conformationally restricted unnatural ligand for immunogen design.

Teaching, Leadership & Public Outreach

Micro-Lectures & Academic Mentorship

National Central University | 2022 – 2023

- Designed and delivered seminars on the foundational physical principles of Nuclear Magnetic Resonance (NMR) spectroscopy for early-career researchers.

Curriculum Design & Youth Science Instruction

New Immigrant Youth Summer Camp (2022 Funded by the National Immigration Agency) | 2019, 2022

- Developed and executed applied chemistry, physics, and ecology curricula translated into vivid experiential workshops for diverse youth.

Community University Lecturer

Community University, Hsinchu City | 2021 – 2022

- Curated public science series titled "The Nobel Laureates Who Reconfigured Society" and "The Amazing Discovery in Microworld," focusing on how science brings changes.

Public Advocacy & Collaborative Communications

Issue-Oriented Social Action Group | 2014 – 2016

- Coordinated educational outreach platforms, reading circles, and analytical workshops investigating power structures in the social system, historical preservation, and contemporary civic policy.

Technical Consultant

Artistic Collaborations (with Artist Ying Chen) | Berlin, Germany | *10/2022 – Present*

- Provide chemical physics and material stability insight to support cross-disciplinary artistic expressions and experimental installations.

Technical Skills

- **Synthesis**: Organic Synthesis, Carbohydrate Chemistry, Chemo-Enzymatic Synthesis, Flexible Protecting Groups, Bioconjugation, Anaerobic Handling.
- Analytical Techniques: 1D/2D Nuclear Magnetic Resonance (NMR), High-Performance Liquid Chromatography (HPLC), Gel-filtration chromatography, LC-MS (ESI), MALDI-TOF mass spectrometry, High-throughput glycan microarray processing.
- **Surface & Materials Characterization**: Transmission Electron Microscopy (TEM), X-ray Diffraction (XRD), Contact Angle Goniometry, Dynamic Light Scattering (DLS), zwitterionic polymer surface grafting.
- **Languages**: English (IELTS 7), German (A1 Certified), Spanish, French, and Japanese.

Selected Honors & Certifications

- **2026:** Advanced Certification in Intercultural Skills for International Scientists – *Gesellschaft Deutscher Chemiker (GDCh), Germany*.
- **2022:** Global Mindset Workshop Selection (International Industrial Collaboration Track) – *STPI Center, NARLabs, Taiwan*.
- **2021:** High-Level Industrial Professionals Cultivation Fellowship – *TIRI, Taiwan*.
- **2020:** Genomics Research Center Bonus Research Award – *Academia Sinica, Taiwan*.
- **2014:** Academic Excellence and Social Service Memorial Scholarship – *Taiwan*.